

Heterogeneous Causal Effects of NYC Congestion Pricing

An Application of Local Linear Causal Forests

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Background and Motivation

- ▶ **Policy Context:** Assessment of the New York City 2025 Congestion Pricing Act on motor vehicle collisions.
- ▶ **Prior Findings:** Initial Difference-in-Differences (DiD) analysis estimated a causal¹ reduction of ≈ 20 accidents per week in Lower Manhattan.
- ▶ **Limitations of ATE:**
 - ▶ Average Treatment Effect (ATE) lacks the granularity required to identify localized policy impacts across different neighborhoods.
- ▶ **Current Objective:** Expand the post-treatment window to 62 weeks and evaluate spatial-temporal heterogeneity in treatment effects.

1: Study had weak parallel trends, limited post-treatment data, and overall poor internal/external validity.

Project Framework

- ▶ **Pivoting to Heterogenous Treatment Effects:** Moving from a global average to a high-dimensional map of policy impact across the congestion pricing zone by mapping NYC as a grid of hexagons.
- ▶ **LLCF Architecture:** Implementation of Local Linear Causal Forests to identify variation across contexts (peak hours, high-density intersections).
- ▶ **Validation of Assumptions:** Testing different spatial sizes of hexagons across NYC, validating parallel trends, assessing the Unconfoundedness principle.
- ▶ **Technical Implementation:**
 - ▶ Solving a Weighted M-Estimation minimization problem for each target prediction point.
 - ▶ Utilization of **Honest Trees** to ensure valid p-values by separating data for structure and estimation.
 - ▶ Bootstrap sub-sampling (infinitesimal jackknife) for robust standard error generation.

Implementation and Timeline

- **Data Augmentation:** NYPD collision reports will be merged with weather data, subway usage rates, and monthly NYC population estimates.



- **Computation:** Use of parallel processing and a high-memory local environment (64GB RAM) for high-dimensional estimation across policy zone.

Timeline	Objective
Weeks 2–3	Data Cleaning, Augmentation, and Revised EDA.
Weeks 4–7	LLCF Derivation, Python Implementation, and Benchmarking.
Weeks 8–11	DiD Comparison and Empirical Paper Finalization.